

SAMARTHI LAHIRI

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📍 Bengaluru

🌐 <https://github.com/Samarthi>

WORK EXPERIENCE

Solutioneering

Software Engineer

June 2023-Present

- Designed and developed AI Agentic data insight generating service that handles 100 RPS using FastAPI
- Designed and developed a scalable autocomplete service using a hybrid architecture of fine-tuned LLM and tries.
- Designed and developed large scale search using Elasticsearch, hybrid-scoring metrics, reranking and click-tracking with 0.98 recall@3, handling 1000 requests per day.
- Improved API performance by ~40% by making codebase async & non-blocking.
- Deployed dockerized monoliths in AWS EC2 instance, with a CI/CD Bitbucket pipeline.
- Solely designed, wrote and owned the backend of diff-tracking web solution for Excel sheets in Python FastAPI.
- Designed modular, maintainable code and wrote exhaustive unit tests.

Intern

Intern

January 2023-June 2023

- Developed a chatbot using Elasticsearch and Python FastAPI.
- Wrote the backend to a web app which lets Pharma affiliate users control versions of their roster and territory spreadsheet files and also marks gaps in the data which reduced manual analyst efforts by about 50%.

Central Drug Research Institute

Blockchain Developer Intern

Jun 2019 - Aug 2019

- Created an MVP from scratch on the Quorum chain using Solidity, and IPFS.
- Worked on a donation tracking and escrow system that held or rescinded donations based on organizational performance of the receiver.

EDUCATION

BITS Pilani

M.E in Software Systems - 7.3 GPA

2021-2023

RCC Institute of Information Technology

B.Tech in Computer Science and Engineering - 8.9 GPA

2017-2021

PROJECTS

Distributed File Drive Manager in Python

- Centralized interface that aggregates and displays all user-stored files from various drive locations
- Implements novel file format to distribute a file over multiple drives if too large to store in a single drive location using protobufs

Ray Tracer in C++

- Ray tracing is an algorithm to mimic the behavior of light rays in the real world to generate ultra-realistic images
- Implements sphere tracing with support for multiple material type

SKILLS

C/C++  RAG  Async Programming 
Data Structures and Algorithms  Python  FastAPI 

- Programming Languages: C/C++, Python, Javascript, Rust
- Databases: MySQL, MongoDB, Neo4j, Elasticsearch
- AWS EC2, Nginx
- Docker, Kubernetes

PUBLICATION

- Cross-chain Transaction Validation using Lock-and-Key Method for Multi-System Blockchain, IEEE International | Conference on Communications Workshops (ICC Workshops), 2023